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Chapter 8
RATIO

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1. In each of the following pairs of measures, find the ratio of the smaller to the larger :

(i) 4 meters, 12 meters (ii) 8 litres, 20 litres

(iii) 20 kilometers, 25 kilograms

(i) 4/12 = 1/3 Ratio = 1:3

(ii) 8/20 = 2/5 Ratio = 2:5

(iii) 20/25 = 4/5 Ratio = 4:5

2. A can contains 4 litres of water and other can, 14 litres

(i) What is the ratio of the amount of water in the smaller can to the amount of water in the larger can?

4/14 = 2/7 Ratio = 2:7

(ii) If one more litre of water is poured into each can, what would this ratio become?

If one more litre of water is poured into each can amount of water in the smaller can is 5 litres and that in the larger is 15 litres

5/15 = 1/3 Ratio = 1:3

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1. The ratio of the inner and outer angles of a regular polygon is 7:2

(i) How much is each angle?

Inner angle = 7X

Outer angle = 2X

7X + 2X = 180

9X = 180

X = 180/9

X = 20

Inner angle = 7 x 20 = 140^0

Outer angle = 2 x 20 = 40^0

(ii) How many sides does the polygon have?

$$\text{Number of sides} = \frac{\text{sum of outer angles}}{\text{one outer angle}}$$

= 360/40 = 9

2. There are four right triangles, all with ratio of perpendicular sides 3:4.
One more information about each triangle is given below.
Calculate the lengths of all three sides of each.

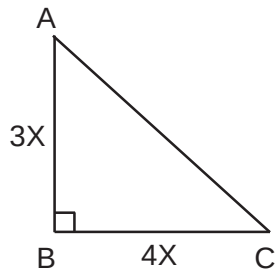
(i) The difference of lengths of the perpendicular sides is 24 meters

Let the perpendicular sides be 3X, 4X

Difference = 24

4X - 3X = 24

X = 24



Hypotenuse, AC $=\sqrt{(3X)^2 + (4X)^2}$

$=\sqrt{9X^2 + 16X^2}$

$=\sqrt{25X^2}$

$= 5X$

Length of perpendicular sides - 3X, 4X

- 3 x 24, 4 x 24

76, 96

Hypotenuse = 5x24 = 120

(ii) The perimeter of the triangle is 24 meters

Permieter = 24

4X + 3X + 5X = 24

12X = 24

X = 24/12 = 2

	3X,	4X,	5X
Sides of the triangle	3x2	4x2	5x2
	6m	8m	10m

(iii) The area of the triangle is 24 square meters

Area = 24

1/2 bh = 24

1/2 x 4X x 3X = 24

$6X^2 = 24$

$X^2 = 24/6$

$X^2 = 4$

X = 2

Sides of the triangle = 3X, 4X, 5X

= 3x2, 4x2, 5x2

= 6m, 8m, 10m

(iv) Hypotenuse 24 meters (Write the lengths in meters and centimeters)

Hypotenuse $5X = 24$

$X = 24/5$

$= 4.8$

Perpendicular sides are $3x4.8$, $4x4.8$

$= 14.4, 19.2$

$= 14m\ 40cm, 19m\ 20cm$

3. The sides of two squares are in the ratio 3 : 4. What is the ratio of their areas?

Sides $= 3X, 4X$

Areas $= (3X)^2 = 9X^2$

$(4X)^2 = 16X^2$

$9X^2 / 16X^2 = 9/16$

Ratio $= 9 : 16$

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1. The length to width ratio of a rectangle is 3:2. The rectangle is enlarged by increasing the length by half much. What is the length to width ratio of the new rectangle?

Length $= 3X$ Width $= 2X$

New length $= 3X + 3X/2 = \frac{6X + 3X}{2}$

$= 9X / 2$

Length to width ratio of new rectangle $= 9X/2 : 2X$

$= 9/2 : 2$

$= 9 : 4$

2. The ratio of 2 angles is 1 : 2. When the smaller angle was increased by 6^0, and the larger angle was decreased by 6^0, this ratio become 2 : 3. What were the original angles?

Angles $= 1X, 2X$

New angles $= X+6, 2X-6$

$X+6 : 2X-6 = 2 : 3$

First angle $X+6$ is $2/3$ part of the second angle $2X-6$

$X+6 = \frac{2}{3} \times (2X-6)$

$3(X+6) = 2(2X-6)$

$3X + 18 = 4X - 12$

$18 + 12 = 4X - 3X$

$30 = X$

Original angles $= X, 2X$

$30^0 \quad 2x30 = 60^0$

3. In a mixture, acid and water are in the ratio 3 : 2. When 15 litres of acid was added to this, the ratio became 3 : 1. How many litres of acid and water are in the current mixture?

Quantity of acid = 3X

Quantity of water = 2X

When 15 litres of acid is added

Quantity of acid = 3X + 15

3X+15 : 2X = 3 : 1

Quantity of acid is 3/1 times of water

3X + 15 = 3 x 2X

3X + 15 = 6X

15 = 6X - 3X

15 = 3X

15/3 = X

5 = X

Quantity of acid in current mixture = 3X + 15

= 3x5 + 15

= 15 + 15

= 30 litres

Quantity of water = 2X

= 2 x 5 = 10

4. The height to width ratio of a rectangle of permiter 54 centimeters is 4 : 5

(i) This ratio is to be made 2 : 3 by increasing the height of width. Which is to be increased? By how much?

(ii) The ratio is to be made 2 : 3 by increasing the height or width. Which is to be decreased? By how much?

Ratio 4 : 5

It means height is 4/5 part of width

If ratio is 2/3, height is 2/3 part of width

$$\frac{4x3}{5x3} = \frac{12}{15}$$

$$\frac{2x5}{3x5} = \frac{10}{15}$$

$$\frac{12}{15} > \frac{10}{15}$$

$$\frac{4}{5} > \frac{2}{3}$$

Height should be decreased

Ratio = 4 : 5

Height = 4X

Width = 5X

Perimeter = 2 x (height + width) = 54

Height + Width = 54/2

Height + Width = 27

4X + 5X = 27

9X = 27

X = 27/9

X = 3

Height = 4 x 3 = 12

Width = 5 x 3 = 15

(ii) Height = 12

Width = 15

Width is 3/2 times of height

Width = $\frac{3 \times \cancel{12}^{6}}{\cancel{2}}$

= 18

18 - 15 = 3

Width should be increased by 3cm

(iii) Height is 2/3 part of width

Height = $\frac{2 \times 15}{3}$

= 10

12 - 10 = 2

Height should be decreased by 2 cm

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1. The angles of a triangle are in the ratio 1 : 3 : 5. What are the angles?

Let the angles of the triangle be X, 3X and 5X.

X + 3X + 5X = 180

9X + 180

X = 180/9 = 20

Angles are 20^0, 3x20=60^0, 5x20=100^0

2. The sides of a triangle are in the ratio 2 : 3 : 4 and the longest side is 20 centimeters longer than the shortest side. Calculate the length of all three sides.

Sides of the triangle are 2X, 3X and 4X.

4X - 2X = 20

2X = 20

X = 10

Sides of the triangle are 2x10, 3x10, 4x10

20, 30, 40

3. The ratio of the length, width and height of a rectangular box is 2 : 3 : 5 and its volume is 3750 cubic centimeters. Find the length, width and the height.

Length = 2X Width = 3X Height = 5X
Volume = 3750
2X x 3X x 5X = 3750
$30X^3 = 3750$
$X^3 = 3750/30$
$X^3 = 125$
$X^3 = 5^3$
X = 5
Length = 2X = 2x5 = 10cm
Width = 3X = 3x5 = 15cm
Height = 5X = 5x5 = 25cm

4. A box contains beads of three colours. The ratio of the number of black beads to the number of white beads is 3 : 5 and the ratio of the number of white beads to the number of red beads is 2 : 3. What is the ratio of the number of beads of all three colours?

Ratio of number of black beads to the number of white beads = 3 : 5
Number of black beads = 3X
Number of white beads = 5X
Ratio of number of white beads to the number of red beads is 2 : 3
Number of white beads = 2Y
Number of red beads = 3Y
2Y = 5X
$Y = \frac{5X}{2}$

Number of white beads = $2Y = 2 \times \frac{5}{2} X$

$= 5X$

Number of red beads = $3Y = 3 \times \frac{5}{2} X$

Ratio of three colours
$= 3X : 5X : 3 \times \frac{5}{2} X$
= 3 : 5 : 15/2
= 6 : 10 : 15

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1. In each of the problems below, the length of a line and the ratio in which a point divides the line are specified. Draw each line and mark the point.

(i) AB = 14cm AP:PB = 3:4
(ii) AB = 12cm AP:PB = 3:5
(iii) AB = 7cm AP:PB = 2:3

(i) AP : PB = 3 : 4
AP = 3/7 x 14 = 6cm
PB = 4/7 x 14 = 8cm

(ii) AP = 3/8 x 12

= 9/2 = 4.5cm

PB = 5/8 x 12

= 15/2 = 7.5cm

(iii) AP = 7 x 2/5

= 14/5 = 2.8cm

PB = 7 x 3/5 = 21/5 = 4.2cm

2. A point divides a 9 centimeters long line in the ratio 2 : 3. Calculate the length of each part.

Length

9 x 2/5 = 18/5 = 3.6cm

9 x 3/5 = 27/5 = 5.4cm

3. Of the two parts got when a line is divided in the ratio 3 : 5, the longer part is 6 centimeters long. What is the length of the whole line?

Ratio = 3 : 5

5X = 6

X = 6/5 = 1.2

Total length = 3X + 5X

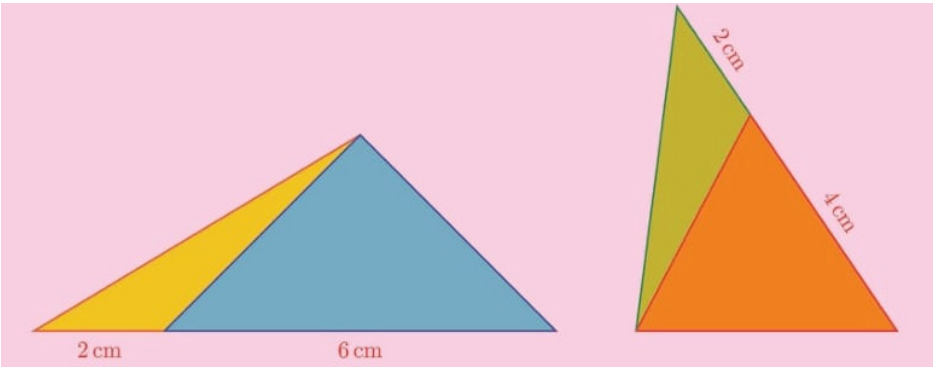
= 8X

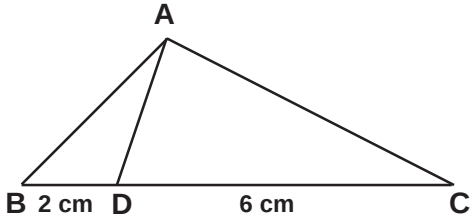
= 8 x 1.2

= 9.6 cm

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1. In each of the two pictures below, a triangle is divided into two smaller triangles. Find what parts of the area of the whole triangle are the areas of the small triangles, in each picture.



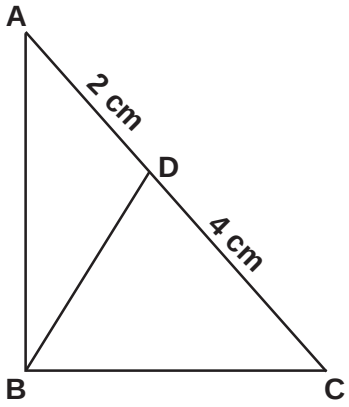


2/6 = 1/3

BD : DC = 1 : 3

Area of the small triangle is 1/4 part of the whole triangle

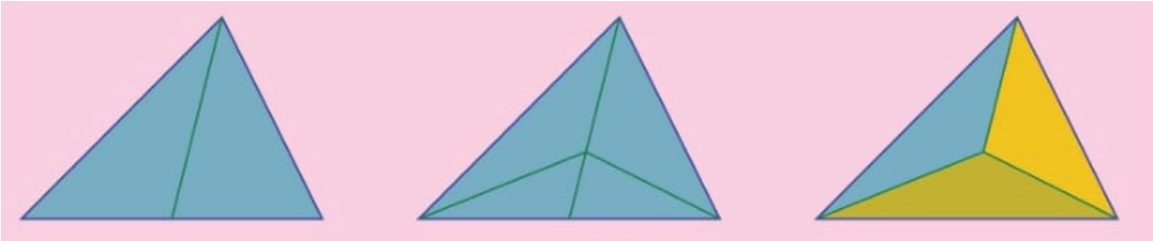
(ii)



AD : DC = 2 : 4 = 1 : 2

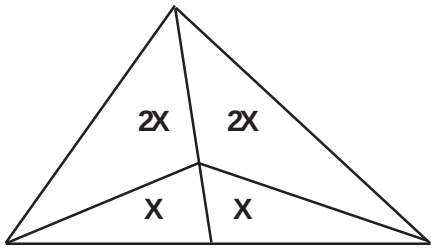
Area of the small triangle is 1/3 part of the whole triangle

2. The top vertex of a triangle is joined to the midpoint of the bottom side, and then the point dividing this line in the ratio 2 : 1 is joined to the other two vertices, as shown in the pictures below:



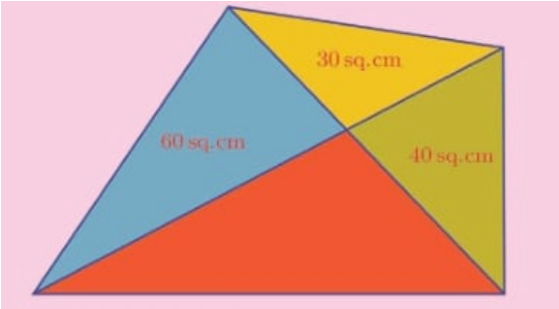
What fraction of the area of the original large triangle are the areas of three small triangles in the third picture?

In the triangle if we join top vertex to the midpoint of the opposite side we get two triangles having half the area of the original triangle. Again if we divide this line in the ratio 2 : 1 the triangle obtained then have that area of the ratio 2 : 1 we can take the area of these triangles as 2X, X

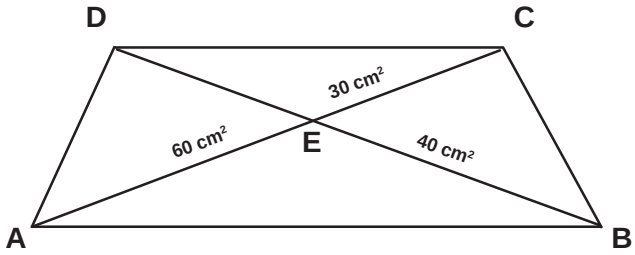


Area of the larger triangle 6X area of each triangle is 2X/6X = 1/3 of the whole triangle.

3. The picture shows a quadrilateral divided into four triangles by its diagonals. The areas of three of these are marked in the picture.



Find the area of the whole quadrilateral



Area of the triangle ADE : Area of △CDE = 60 : 30 = 2 : 1
AE : EC = 2 : 1
Area of △ABE : Area of △BCE = 2 : 1
Area of △BCE = 40
Area of △ABE = 2 x 40 = 80cm^2
Then the area of the quadrilateral = 60 + 30 + 40 + 80 = 210cm^2

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